Lesson Plan · Thursday, November 12

Algebra I · Mr. Pandey · Module 3, Topic 1, Lesson 3: Modeling Linear Inequalities (Day 2 of 3) · essentials only (✪)

Objectives (students will…)

- Solve two-step linear inequalities algebraically.
- Round answers sensibly in context (whole boxes).
- Explain why dividing by a negative reverses the inequality sign.

TEKS

- **A.5B** solve linear inequalities in one variable
- **A.2C**
- Process: A.1A

Key terms

- solve an inequality

Materials

- Slides: m3t1-lesson3-modeling-linear-inequalities (.html/.pptx) · publisher slides with answers: BLMathematics_AlgI_M3_T1_L3_LessonSlides_ENG_with_answers.pptx
- Bell work: BELL WORK/T1-Bell-Work-Thu-11-12.pdf + key (online: m3t1-lesson3-bellwork-day2)
- Exit ticket: EXIT TICKET/T1-Exit-Ticket-Thu-11-12.pdf + key (written like the End of Topic 1 test)
- Video: videos/M3T1-Lesson3-Linear-Inequalities.mp4 · IEP and 504 versions of all of these: IEP/, Section 504/, and the -IEP / -504 PDFs
- Student Edition pp. 376–379 (Activities 3.2, 3.3)
- Inequality-words chart from Wednesday
- Whiteboards

Bell work (on the board)

- What value of x makes $2x + 5 = 17$ true? C. 6
- Which describes the graph of $x \geq 2$ on a number line? J. closed circle at 2, shaded right
- Is $x = -5$ a solution of $-x > 4$? A. Yes, because $5 > 4$
- Chloe buys 2 muffins for $1.50 each and some juices for $2.25 each. Which equation gives the total cost, y, for x juices? G. $y = 2.25x + 3$

Agenda (60 minutes)

Time	Activity	What to do / say
0–5	Bell work	Q4 previews today: the sign flips when the x has a negative coefficient.
5–25	✪ Activity 3.2: Solving Two-Step Linear Inequalities · SE pp. 376–377	Define *solve an inequality*. Worked Example: $3.75b + 25 \geq 1100 \rightarrow b \geq 286.6\ldots \rightarrow$ Q1 at least 287 boxes. Q2a still under $600: $b < 153.3\ldots$, at most 153 boxes. Q2b at least $1500: $b \geq 393.3\ldots$, at least 394 boxes. Q3: same steps as an equation, whatever the form.
25–50	✪ Activity 3.3: Reversing Inequality Signs · SE pp. 378–379	Hike from 4800 ft down 20 ft/min: Q1 $h(m) = -20m + 4800$. Q2 slope -20, y-int 4800, x-int 240 min, domain $0 \leq m \leq 240$. Q3 graph: below 3200 ft → $m > 80$. Q4 algebra: $-20m + 4800 < 3200 \rightarrow -20m < -1600 \rightarrow m > 80$ (flip!). Q6 table: every row's sign reverses. **Board rule:** multiply or divide by a negative → flip the sign.
50–58	Exit ticket	1. What is the solution set for $-3x + 4 \geq 19$? $x \leq -5$ 2. Solve and graph: $5 - 2x < 11$ $x > -3$, open circle at -3, shaded right 3. Tire: 26 psi, −0.1 psi/min, stay ≥ 22. Solve for t. $t \leq$ 40 minutes IEP/504: read aloud on request, properties/notes card, extra time.
58–60	Wrap up	Homework: Lesson 3 Assignment Practice 1e–g and 2a–b (SE pp. 384–385).

Check for understanding

- Whiteboards: "$-2x < 8$. Answer?" $x > -4$
- Thumbs: "Subtracting a negative flips the sign." Down: only × or ÷ by a negative
- Activity 3.3 Q1–7 are the TE's check points.

After class

- Forgot to flip (exit Q1–2) → Skills Practice Set C.
- Missed context Q3 → pair with a partner on Monday's bell work.
- All correct → Skills Practice Set D.

3.2 and 3.3 are starred. Rounding in context: "at least" rounds up (287), "at most" rounds down (153).